

EDUCATION

M.S. in Computer Science, Michigan Technological University, GPA: 3.85/4.0

Aug 2023 – Apr 2025

B.Tech in Computer Science, University of Engineering & Management, GPA: 3.39/4.0

Jul 2016 – May 2020

WORK EXPERIENCE

Cognizant

Jan 2020 – Jun 2023

Client: Capital One

Software Engineer II

- Architected and launched a Fraud Detection Microservice using Spring Boot REST APIs on AWS, with P95 ≤ 150 ms latency and 99.95% uptime, serving 2M+ users and ensuring compliance with financial regulations.
- Proactively identified and engineered fault-tolerant Kafka pipelines by implementing idempotent producers, exponential backoff retries, and consumer deduplication, resolving delayed and duplicate notifications and reducing end-to-end latency by 45% and duplicate incidents by 95%.
- Eliminated manual deployments by architecting a Jenkins CI/CD pipeline, integrated with AWS CodeDeploy, IAM, and Lambda notifications, automating build, test, and release validation, and reducing deployment errors by 99%.
- Implemented IaC governance for Kubernetes and Terraform (policy checks and gated CI), blocking insecure configs before deployment and reducing security/ops exceptions by 40% while improving deployment success to 99%.
- Containerized React/Next.js frontend and Java services with Docker and orchestrated deployments on AWS EKS using Helm charts and ArgoCD (GitOps), allowing blue/green, rolling, and canary releases, cutting release cycles by 50%.
- Mentored 5 interns, led code reviews, and introduced shared coding standards adopted across 3 teams, reducing post-release defects by 45% over 6 months by eliminating inconsistent patterns causing recurring production issues.

Client: Walmart

Software Engineer I

- Enhanced advanced filtering, sorting, and pagination across a catalog of 5M+ product listings, reducing server response time by 20% and database load for 50M+ monthly queries, improving product discovery performance at scale.
- Owned observability for Java and Kafka services via CloudWatch and Splunk, built dashboards, alerts, and runbooks, and led RCA follow-ups, cutting MTTD by 40%, MTTR by 25%, and defects by 20% across Sev-2/Sev-3 incidents.
- Redesigning product listing pages, transactional emails, and shopping cart UI using React, TypeScript, and Tailwind CSS, reducing customer support inquiries by 15% through clearer communication and streamlined workflows.
- Resolved frontend and backend production issues through root cause analysis, reducing downtime from 12 hours per quarter to 30 minutes and cutting average production incidents from 250 to 20 per month.
- Built and maintained unit, integration, and end-to-end test suites using JUnit, Mockito, Jest, and Playwright, following Test-Driven Development (TDD) practices, increasing test coverage from 45% to 90% and cutting production regressions by 60%.

CURRENT RESEARCH EXPERIENCE

Michigan Technological University, Research Assistant

Jun 2025 – Present

- Built and iterated AI/ML prototypes using Python, translating research ideas into working systems and validating real-world applicability across multiple experimental runs and datasets.
- Designed evaluation frameworks and KPI-driven benchmarks to measure model performance (accuracy, latency, reliability), improving experiment comparability and decision-making speed by 35%.
- Developed automated data pipelines for experimentation and analysis, reducing manual preprocessing and evaluation effort by 40% and accelerating iteration cycles.
- Collaborated with researchers and stakeholders to define problem statements, analyze large-scale datasets, and deliver insights that influenced research direction across multiple projects.

SKILLS

Languages: Java, SQL, Go / Golang, Python, TypeScript, JavaScript

Backend: Spring Boot, Spring MVC, Spring Data JPA, Spring Security, REST APIs, GraphQL APIs, Node.js

SQL/NoSQL Databases: PostgreSQL, Cassandra, Redis, Elasticsearch, MongoDB

Observability & Event Streaming: AWS CloudWatch, Apache Splunk, Apache Kafka, RabbitMQ

Cloud/DevOps: AWS (EC2, EKS, S3, Lambda, CodeDeploy, IAM, CloudWatch), Docker, Kubernetes, ArgoCD (GitOps), Terraform, Helm, Jenkins, Maven, Git, Linux

Testing: JUnit, Mockito, Jest, Playwright

Frontend: Angular, Next.js, React.js, Redux, Zustand, Tailwind CSS, HTML5, CSS3

TECHNICAL PROJECTS

Meet AI | [Demo](#) | [Live Site](#) | [GitHub](#)

- Built a live AI agent that joins video calls using OpenAI's Realtime API, delivering sub-second voice responses with a natural conversational feel.
- Eliminated manual mid-meeting searches by enabling direct agent Q&A, reducing answer retrieval from 3–5 minutes to under 3 seconds.
- Designed agent-based orchestration with Inngest Agent Kit to automate multi-step post-meeting workflows.
- Built a pipeline combining transcript fetching, parsing, speaker enrichment, and LLM-based summarization.
- Engineered dynamic system prompt generation, adapting per meeting context, agent instructions, and conversation state.
- Built with Next.js, Ngix, OAuth, JWT, Tailwind CSS, WebRTC, tRPC, Drizzle ORM, and Polar API.

Plant Disease Detection | [Demo](#) | [GitHub](#)

- Built a full-stack AI application (React.js and Python/FastAPI) to classify leaf images as Early Blight, Late Blight, or Healthy, enabling sub-2s end-to-end predictions for non-technical users.
- Designed and trained Python-based CNN models in TensorFlow/Keras, experimenting with 3+ architectures (custom CNN, and transfer learning), and achieved the best test accuracy of 95%
- Built an end-to-end Python ML pipeline (data to model to REST API to web UI), cutting model iteration time by 50% through reusable preprocessing and repeatable training runs.
- Built a Python FastAPI inference endpoint delivering single-image predictions in 150 ms (p95) with top-1 label and confidence, deploying a lightweight exported Keras (.h5) model optimized for low-memory and edge inference.